

FINA4370A Derivatives Trading: Analysis and Strategies

Unit 4: Understanding the Volatility Surface

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August 9, 2026

Outline

- 1 Introduction: Beyond a Single Number
- 2 Implied vs. Realized Volatility
- 3 The Volatility Skew (or Smile)
- 4 The Volatility Term Structure
- 5 Strategic Implications
- 6 Appendix: Visualizing Volatility in IBKR

Beyond Constant Volatility

- The Black-Scholes-Merton (BSM) model assumes a single, constant volatility for all options on an asset.
- **Reality Check:** This is not true. If we solve for the volatility that matches an option's market price, we get different values for different strikes and expirations.

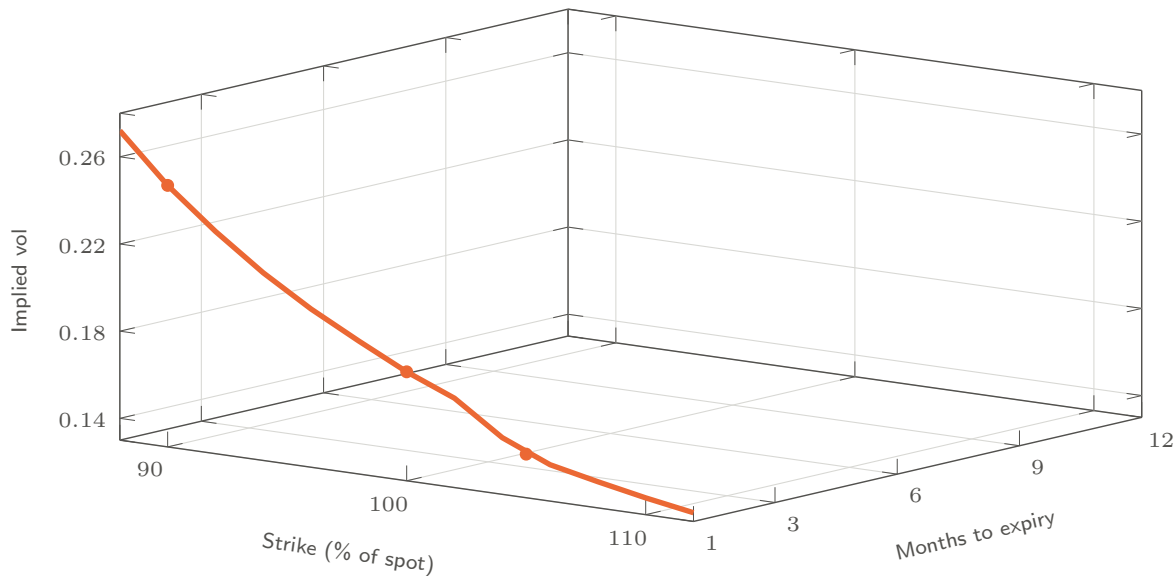
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- This calculated value is the **Implied Volatility (IV)**.
- When we plot these IVs, a three-dimensional picture emerges: the **Volatility Surface**.
- This surface is a map of the market's expectations and is critical for sophisticated trading.

The Whole Object, Before We Slice It



One number per (strike, expiry). The orange edge is the front month — Example 4.1 lives on it. The next two slides cut this object twice: across strikes at a fixed expiry (the **skew**), and along expiry at a fixed strike (the **term structure**). The skew is steep at the front and nearly flat at a year: it decays roughly as $1/\sqrt{T}$.

Implied vs. Realized Volatility

Realized Volatility

- Also called historical volatility.
- The actual, measured volatility of past returns.
- It is backward-looking and factual.
- "How much *did* the asset move?"

Implied Volatility

- The market's consensus forecast of future volatility.
- It is forward-looking and derived from option prices.
- "How much *does the market think* the asset will move?"

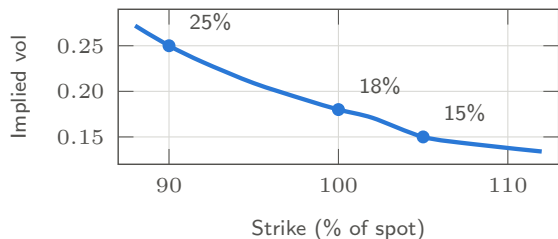
The Volatility Risk Premium

In most markets, Implied Volatility tends to be higher than subsequently Realized Volatility. This difference is the **volatility risk premium**. Option sellers earn this premium as compensation for taking on crash risk.

The Volatility Skew / Smile: A Slice Across Strikes

The volatility skew / smile describes how implied volatility changes as the strike price changes for a **single expiration date**.

- If the BSM model were perfect, this would be a flat line. It is not.



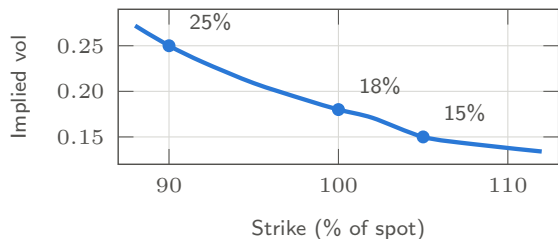
Equity index — monotone downward skew

- IV is highest for low-strike (OTM) puts.
- Slopes downward as strike increases.
- Implies market fears a large *downward* move more than a large upward move.

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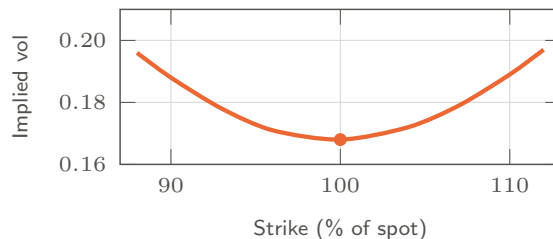
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Currency pair — symmetric smile

- IV is lowest for at-the-money (ATM) options.
- Rises for both OTM puts and OTM calls.
- Implies market fears a large move in *either* direction ("fat tails").

What the Skew Actually Says

One-month options, index at 100, $r = 4\%$. **90% strike put: IV 25%. ATM: 18%. 105% strike call: 15%.**

Price the wings against a flat 18%

	Own IV	Flat 18%
90 put	\$0.198	\$0.033
105 call	\$0.335	\$0.559

Downside protection is **6.1×** dearer; upside participation **40% cheaper.**

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Translate into probabilities

	Skew	Flat
$P(S_T < 90)$	7.08%	1.94%
$P(S_T > 105)$	14.20%	18.38%

A 10% fall is priced **$3.6\times$ more likely** than lognormal; a 5% rally *less likely*.

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Fat left tail, thin right tail. Every strategy that **buys** downside protection is expensive here; every strategy that **sells** it is well paid.

Why Does the Equity Skew Exist?

The downward skew in equity markets is driven by fundamental factors.

1. Crash Risk (Fear)

- The primary driver. Investors fear a repeat of events like the 1987 crash.
- This creates high demand for OTM puts as portfolio insurance, driving up their prices and thus their implied volatility.

2. Leverage Effect

- As a stock's price falls, its debt-to-equity ratio increases, making the company riskier.
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3. Supply and Demand

- Structural demand for OTM puts from institutions.
- Structural supply of OTM calls from covered call writers.

The Volatility Term Structure: A Slice Across Time

The term structure describes how implied volatility changes as time to expiration changes for a **single strike price** (usually ATM).

Upward Sloping (Contango)

- Short-term IV < Long-term IV.
- This is the **normal** state in calm markets.
- Reflects general uncertainty about the distant future, but no immediate, specific threats.

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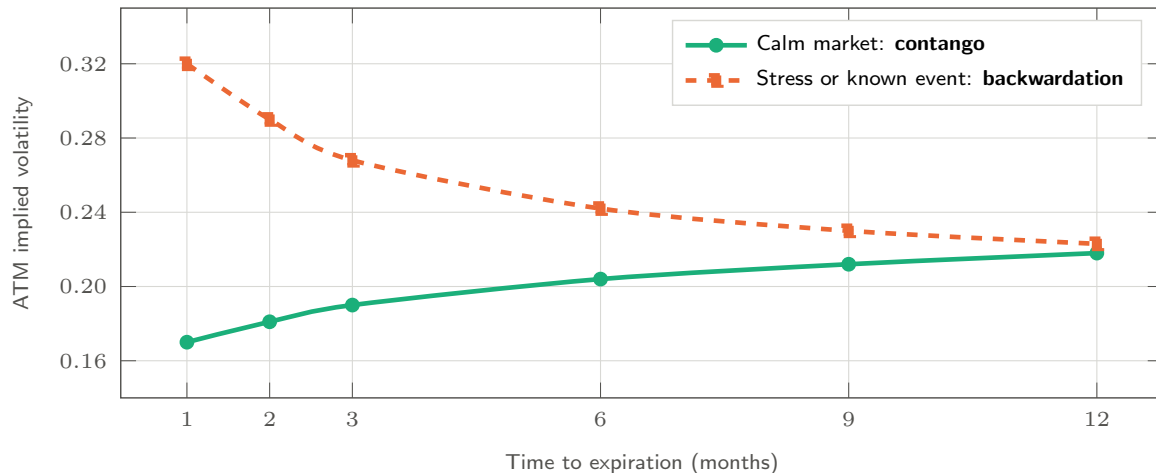
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Downward Sloping (Backwardation)

- Short-term IV > Long-term IV.
- Common during **market stress** or before a major known event (e.g., earnings announcement).
- The market is pricing in a significant near-term risk.

The Term Structure, Drawn



Both curves converge at the long end. The near month moves; the one-year barely does. That is why a calendar spread is a trade on the *front* of the curve, and why the 1–3 month segment is where an event repriced.

Trading the Skew: Risk Reversals

A view on the skew is a view on the relative price of OTM puts vs. OTM calls. The purest way to trade this is with a **risk reversal**.

The Risk Reversal

A position of buying an OTM call and selling an OTM put (or vice-versa) with the same expiration. The price of a risk reversal is a direct market measure of the skew's steepness.

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Example: Trading a Steepening Skew

- **View:** The market is too complacent; fear will increase, causing the skew to steepen (OTM put IV will rise relative to OTM call IV).
- **Strategy:** Buy an OTM put and simultaneously sell an OTM call.
- **Outcome:** This position profits if the skew steepens as predicted. It also has a negative delta, profiting if the underlying asset falls.

The Risk Reversal — and Why It Is Not a Pure Skew Trade

Buy the 25-delta put, sell the 25-delta call, on the same surface:

	Strike	IV	Price	Delta
25-delta put	96.6	20.38%	\$0.904	−0.250
25-delta call	103.6	15.85%	\$0.667	+0.250

$$\text{Risk reversal} = 20.38\% - 15.85\% = +4.53 \text{ vol points}$$

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Why 25-delta? Vega is $S\sqrt{T}\varphi(d_1)$ with φ even, so matched $|\text{delta}|$ forces matched vega: $|d_1| = 0.6745$ on both legs, **\$0.0917 per vol point** each. A parallel shift **cancels**; a steepening **adds** on both — so the skew P&L is the *full* leg vega, not half.

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*Locate on option
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even function ($\therefore$ bell shape)

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Now look at the delta

Net delta = $-0.250 - (+0.250) = \underline{-0.500}$. A one-point fall makes \$0.50 — more than *twice* the entire \$0.237 premium. A one-vol-point steepening, the actual bet, is worth about \$0.092.

The spot exposure is about five and a half times the skew exposure. To trade the skew, buy 50 shares per 100 risk reversals.

Trading the Term Structure: Calendar Spreads

A view on the term structure is a view on the timing of volatility. The classic strategy is the **calendar spread**.

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Example: Trading a Volatility Collapse

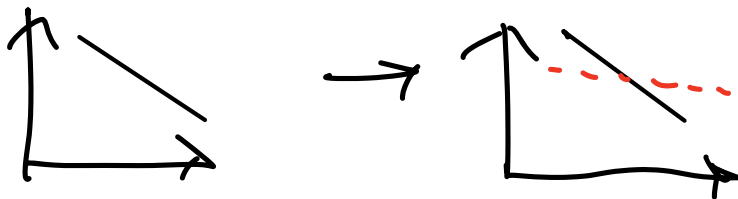
- **Scenario:** Term structure is in steep backwardation before an earnings announcement (1-week IV=80%, 3-month IV=30%).
- **View:** Post-announcement, the 1-week IV will collapse.
- **Strategy:** Sell the expensive short-dated straddle and buy the cheaper long-dated straddle.
- **Outcome:** Profits as the short-dated IV falls sharply relative to the long-dated IV.

Trading the Term Structure — and Being Right Anyway

Index at 100, $r = 4\%$, two weeks before a macro announcement. 1-month IV **32%**, 3-month **24%** — backwardation. Sell the 1-month ATM call (\$3.819), buy the 3-month (\$5.235). **Debit \$1.416.**

Two weeks later the curve has normalised to 1m 18%, 3m 21%:

Index	Short leg	Long leg	Spread	P&L
Unchanged at 100	\$1.591	\$4.235	\$2.643	+\$1.227 (+87%)
Gaps up to 112	\$12.176	\$13.319	\$1.143	-\$0.273 (-19%)
Falls to 92	\$0.019	\$1.132	\$1.113	-\$0.303 (-21%)



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Debit \$1.416. *Introduce Non-linearity $\Rightarrow$ Care about Gamma*

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Right on the thesis in all three rows — and losing in two

A calendar is long vega and **short gamma**. It needs the curve to normalise *and* the index to stay near the strike. **A relative-value trade can be right and still lose, because the exposure it was not designed to take was left unhedged.**

The Discipline Behind Both Trades

Reading a relative-value trade correctly

A structure is only a “pure” expression of a view once the exposures you are **not** trying to take have been hedged away.

- A **risk reversal** trades the slope of the skew — delta-hedged.  $\rightarrow \Delta / \text{Gamma}$
- A **calendar spread** trades the term structure — delta-hedged.  $\rightarrow$
- A **dispersion trade** trades correlation — vega-weighted.  $\rightarrow$

Otherwise the P&L is dominated by direction, and the trader concludes they were right or wrong about the surface **on evidence that has nothing to do with it.**

How the Surface Moves: Sticky Strike vs Sticky Delta

Everything so far describes the surface *at a point in time*. A strategist also needs a view on **how it moves when spot moves** — because that determines the true delta of every position.

Sticky strike

The IV attached to each **strike** stays put as spot moves. The Black–Scholes delta is then correct as computed.

Sticky delta (moneyness)

The IV attached to each level of **moneyness** stays put — the whole surface slides sideways with spot.

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The sign is the opposite of what most people guess

$\Delta_{\text{eff}} = \Delta_{BS} + \mathcal{V} \partial\sigma/\partial S$. Under **sticky delta**, a fall in spot makes a fixed strike *less* OTM, so it slides *down* the skew and its IV **falls**: $\partial\sigma/\partial S > 0$, and a long put's effective delta **shrinks**. On a 95 put with the index at 100, $\Delta_{BS} = -0.159$ but $\Delta_{\text{eff}} = -0.139$ — hedging on model delta **over**-hedges.

→ Delta from Black–Scholes

The Third Regime — the One People Actually Mean

Regime	$\partial\sigma/\partial S$	Long put's effective delta
Sticky strike	0	model delta is correct
Sticky delta	> 0	shrinks — model delta <i>over</i> -hedges
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Only the level shift produces the effect the folklore attributes to sticky delta. If you cannot say which of the three you assume, you cannot say what your delta is.

Appendix: Visualizing Volatility in IBKR (optional)

Interactive Brokers TWS provides professional-grade tools to analyze the volatility surface in real time.

The Implied Volatility Viewer

The primary tool for analyzing the skew and term structure is the **Implied Volatility Viewer**.

- In Mosaic, click **New Window** and select **Option Analysis** → **Interactive Analysis** → **Implied Volatility Viewer**.^a
- It plots implied volatility (Y-axis) against strike price (X-axis).

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What this tool shows you:

- **The Skew:** The shape of the curve for a single expiration is the volatility skew.
- **The Term Structure:** Each line on the plot represents a different expiration date. By comparing the lines, you can see if the term structure is in contango or backwardation.

The 3D Volatility Surface Plot

For a complete picture, TWS offers a 3D visualization of the entire volatility surface.

- Click the **3D** button on the Implied Volatility Viewer; the surface can then be rotated and zoomed.
- It provides an intuitive way to see the interaction between the skew and the term structure.

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Strategic Use Cases

- Identify the steepest part of the skew to find expensive puts to sell.
- Spot kinks or irregularities that might represent mispricings.
- Visually confirm the term structure shape across all strikes.